

Intellectual property management: A knowledge supply chain perspective

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With greater reliance on outsourcing has come a rise in the importance of supply chain and knowledge management principles. Intellectual property management in particular has gained saliency. Licensing relationships are like those between buyers and suppliers, except that the former transact knowledge while the latter transact concrete materials and services. This analogy gives rise to five licensing models: stick (damage recovery), defense (posturing), carrot (attracting potential suitors), consortium (standard setting), and market (industry-wide usage). Adding to the literature, these models can be used as different lenses through which to view licensors/licensees in the area of IP management.

Business in the United States has undergone dramatic changes over the last two decades. Companies have been reengineered, downsized, and restructured. Employees have been dismissed, divisions scrapped, and subsidiaries spun off. Consequently, many managers have found ways to work with less in order to maintain the same level of output. As part of the natural evolution of changing competitive strategies, companies have increased their outsourcing to external suppliers, resulting in operational and corporate achievements becoming more closely integrated and intertwined with those of the suppliers.

Out of these trends, two managerial issues have gained saliency: supply chain management and knowledge management. In the past, handling the supplier interface was a task relegated to purchasing function specialists; today it is discussed by top executives and referred to as *supply chain management*. Buyer/supplier relationships have received ample attention in the management arena, and are bound to receive more as more manufacturing is delegated to suppliers and contract manufacturers. Because of this, companies have had to search for new competitive advantages within *knowledge management*. Consequently, intellectual property (IP) management has taken on new strategic meaning because of its critical impact on corporate success.

As shown in **Table 1**, whereas a typical buyer/supplier relationship involves the delivery of materials and services from the supplier to the buyer, knowledge is the basis of the transaction in a licensing relationship. Licensors are delivering knowledge to the licensees. Thus, the supply chain involving IP is referred to as a *knowledge supply chain*. In this context, we define knowledge as technologies, inventions, and know-how that help businesses bring products to market. The material flow is like the physical flow of electrical parts; the knowledge flow is like the flow of techniques that weld these parts together.

Moreover, payment in a buyer/supplier relationship is typically made for each shipment delivered to the buying

Table 1
Buyer/supplier relationship versus licensor/licensee relationship

<i>Managerial issues</i>	<i>Buyer/supplier</i>	<i>Licensor/licensee</i>
Transacting item	Materials and services	Knowledge
Sender of transactions	Supplying company	Licensing company
Terms of payment	Per delivery	Per agreement
Major players	Buyers and engineers	Lawyers and business managers
Length of relationships	Short-term and long-term	Generally long-term

patent owner can prohibit the sale, manufacturing, or distribution of products that embed the IP by court injunction, and can receive monetary rewards as compensation for damages. The licensor uses this power to strike an agreement with a second party, who becomes a licensee once the agreement is made. Hence, a relationship is established and a knowledge supply chain formed.

company, where major players are purchasing professionals and engineers. However, in the knowledge supply chain, payment is based on the agreement drafted in a legal contract, where lawyers and business managers play a major role.

Moreover, depending on the type of items being purchased, the length of the relationship varies among buyers and suppliers. For instance, if the item is a commodity purchased in an open market, the relationship is likely to be short-term, even a one-time transaction; if the item involves design and engineering, the relationship tends to stretch over several years. However, with licensing, the relationship tends to last as long as the IP remains current and valued in the market. Once it is established, it presumably endures over a long period of time, because severing it would require the daunting task of separating out the various aspects of knowledge embedded in product design or manufacturing methods—often layers upon layers of products and processes. For example, a specific technology itself must be teased out of a product, a task made more difficult by the fact that the transfer from the licensor to the licensee involves technology modification in order to yield a manufacturable and profitable product for the licensee.

Intellectual property and knowledge supply chains

Intellectual property comes primarily in the form of patents, trade secrets (know-how), trademarks, and copyrights. Here, we focus on patents and trade secrets because they are key IP assets in high-technology industries. Patent laws can provide a legal barrier for the manufacture or sale of products containing particular knowledge, even though the knowledge is publicly disclosed to promote more knowledge creation. Trade secrets erect barriers by guarding knowledge from potential competitors. Licensing agreements offer a means for two companies to overcome both of these barriers and work together in a relationship.

Licensing ensures the fair treatment of IP for both licensor and licensee. It entails an agreement between the two to “hold harmless” to each other. The licensor’s knowledge is protected by a patent, which precludes a second entity from making, using, selling, or importing the IP. The

Such relationships often involve the licensor’s assistance and training to ensure that the licensee can use the knowledge effectively.

In the new era of competing through knowledge, the issue of how to manage the flow of IP in and out of a company has become a major corporate concern. The licensing that controls this knowledge flow has become a huge industry. It is also lucrative, as shown by companies such as IBM, which now licenses any technology it has developed. Alternatively, infringement of IP rights owned by others may lead to severe financial consequences. Avant! paid about \$200 million to Cadence in an infringement suit for its trade secret theft of integrated circuit design automation software. Establishing and managing this particular type of knowledge supply chain, therefore, requires several different models of voluntary and in-voluntary licensing relationships. This is where our model framework comes in.

The five models of licensing relationships

Many researchers in supply chain management focus on the buyer/supplier relationship. Our focus is on the licensor/licensee relationship as we bring together the issues of supply chain and knowledge management. Here we introduce five basic models of this relationship, which have strategic implications for managing IP from both licensor and licensee perspectives.

Summarized in **Table 2**, these five models are by no means an exhaustive set of licensing relationships. They are, nonetheless, the most dominant forms based on our experiences and the review of literature in this area. Each model is accompanied by a pictorial rendition of the relationships among the involved companies. Large circles represent larger, dominant companies, while small circles represent smaller firms. The arrows indicate the direction of the primary force for establishing the relationship. Each model’s discussion ends with a case from the semiconductor industry as an illustration. We chose the semiconductor industry because of the cumulative nature of its knowledge and the importance of sharing that knowledge.

Table 2
Licensing relationships: Five models

<i>Model</i>	<i>Nature of relationship</i>	<i>Licensor's main motivation</i>	<i>Licensee's main motivation</i>	<i>Knowledge flow</i>	<i>Money flow</i>
Stick	Adversarial	To prevent infringement of IP assets	None	Directional	Directional
Defense	Mutual	To avoid unnecessary conflict	Same as licensor's	Bi-directional	Directional based on asset differentials
Carrot	Strategic	To capitalize on existing knowledge	To apply knowledge to product	Directional	Directional
Consortium	Collaborative	To create industry standards	Same as licensor's	Multi-directional	None
Market	Coerced	To embed technology in new products	To increase compatibility	Directional	Directional

Its technology is a cumulative system technology in which inventions of many companies overlap. The complexity of its products and manufacturing processes is simply too great to be developed by a single company. Sharing knowledge via patents and trade secrets enables companies to participate in the market and compete.

The stick model

A company with a dominant IP position, holding a generous patent portfolio of various types of knowledge, controls who becomes part of its knowledge supply chain. As shown in **Figure 1**, it reaches out to other companies and uses legal mechanisms to extract damages, impose royalties to compensate for R&D investment, and maintain a competitive edge. By so doing, it tries to fend off competitors and market entrants by aggressively policing them.

Many high-tech firms with revenues of a billion dollars or more have significant R&D investments and broad patent portfolios. Their management is now responsible for extracting value by focusing on tangible gains from damages or settlement charges, or intangible gains from future options. According to Hall and Ziedonis (2001), companies that have historically used the stick model include Texas Instruments, AT&T, IBM, and Motorola. However, their approaches for damage recovery often differ.

A company carries a "stick" if it finds evidence of its patents being infringed upon, and if it has sufficient funds to prosecute the allegedly infringing firm. In many cases, relative business size and patent portfolio strength provide the keys to a successful prosecution. Defending the return on R&D, commercialization, and anticipated economic damages are only a few of the obvious reasons for such aggression. Building a portfolio of related patents helps the company stake claims in current

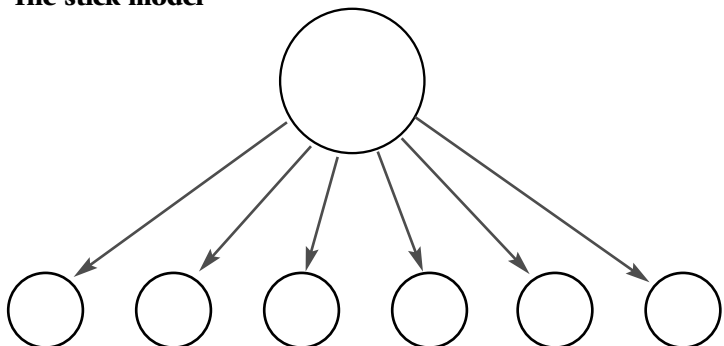
or future markets. Depending on the strength of the patent portfolio and, in particular, of the allegedly infringed patent, the conflicting situation may find resolution in monetary compensation. Many cases settle out of court, ending in royalty payments to the patent rights owner or some type of agreement that will ensure equity in the knowledge supply chain relationship.

The competitive intelligence of potential infringers and expert patent lawyers to support infringement claims is necessary for patent holders. Big companies such as Texas Instruments have extensive legal departments that smaller companies cannot afford; little guys must resort to outsourcing on a case-by-case basis. Therefore, the stick model is generally pursued by companies with high capitalization.

● The case of Texas Instruments

In 1981, facing growing competition from Asian electronics manufacturers, Texas Instruments decided to aggressively enforce patent rights on handheld calculators. The target was Casio, one of Japan's major electronics produc-

Figure 1
The stick model



ers with a large foothold in Asia. Casio battled for over a year to avoid having the case heard in Texas. But when it became clear that a Texas jury would decide the case, Casio signed the licensing agreement on TI's terms. By using the Casio settlement as a model, reports Warshofsky (1994), TI licensed the rest of the Japanese calculator industry.

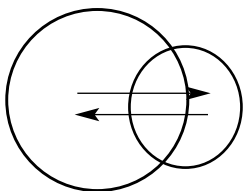
TI's next major battle was a series of highly publicized lawsuits against dynamic random-access memory (DRAM) products in Japan and Korea. In 1985, TI sued nine companies: Hitachi, NEC, Mitsubishi, Matsushita, OKI, Fujitsu, Toshiba, Sharp, and Samsung. The successful settlement of the DRAM cases brought TI over \$1.8 billion in royalties between 1986 and 1993. It also enabled the company to maintain a relatively high level of R&D spending, especially during the 1989–1991 industry downturn. Finally, TI's aggressive application of the stick model had a major positive impact on enforceability of patent rights outside the United States, and led to multiple cross-licensing arrangements between American and Asian companies. According to Grindley and Teece (1997), Asian companies recognized the strategic importance of IP assets, began investing more to strengthen their patent portfolios, and heightened their awareness of the importance of IP rights.

The defense model

The patent licensing practices of major semiconductor companies also focus on freedom of action through forward-looking cross licensing. In the stick model, motives for entering into a cross-licensing arrangement were based on value extraction from R&D through past damages and royalty payments. Going forward, these arrangements have very important implications for companies' behavior and technological progress.

Knowledge, especially in the context of IP, can become a cashless equity. While being kind to the balance sheet, a mutual agreement not to sue each other allows each company to continue doing business in its respective market while using some aspect of the other firm's technology. But such mutual agreements often rely on comparable portfolios. If one company's portfolio stands at a higher equity level, then an agreement about the royalty amount can be reached. This is where the defense model comes in.

Figure 2
The defense model



The larger circle in **Figure 2** symbolizes the company with the more extensive portfolio, and the arrows point in both directions to indicate that the desire for a relationship is mutual. In other words, the licensing relationship is reciprocal in the knowledge supply chain. This

defense model is practiced by companies entering into a cross-licensing arrangement with the objective of securing "freedom of action" in a particular industry or segment.

Such reciprocal exchange of knowledge may even occur between major industry rivals. By leveling the technological playing field, both firms can avert astronomical legal fees and focus better on their core competencies and marketing programs. For companies such as Advanced Micro Devices, Intel, TI, and Vanguard International Semiconductor, practicing the defense model means driving innovation faster than the competitor. Because the average patent granting process takes about two and a half years, most of the bleeding-edge technology would already be outdated. Adopting this defense model allows participants to introduce new technologies and products without fear of infringing IP rights owned by others, thereby limiting litigation exposure and associated costs. It can also generate revenue for companies with stronger IP portfolios. The key issue revolves around determining the parity of portfolios, which very often include thousands of patents. If the companies are unable to reach parity, they must at least agree over who owes what and how much.

As one might imagine, the downside of practicing the defense model is its complexity. After a while, the entangled relationships that emerge from multiple cross-licensing agreements become a veritable labyrinth. Complexities increase because, depending on the scope of the agreement, companies with large portfolios can take a considerable amount of time to evaluate and arrive at a fair assessment.

Strong acquisition and divestiture activities of recent years create further complications. When a portion of the licensed agreement is taken away through sales, even to a competitor, the previously achieved balance of portfolios and royalties is disturbed and requires re-evaluation. Fortunately, most contracts provide protection for at least some number of years, allowing companies to transfer technology from one source to another.

● The case of RCA

A patent is essentially the right to exclude. But in cumulative system technologies, the desire to exclude other companies from using a particular invention can lead to a deadlock. The history of the commercialization of radio gives us an excellent example of both the challenges and solutions found in cross-licensing arrangements.

Radio commercialization required the use of many inventions and associated patents. But the individuals and corporations that owned the patents initially refused to enter into cross-licensing arrangements. This reluctance interfered with commercialization efforts and required a solution. In the United States, the situation was resolved under pressure from the US Navy, which needed to harness the available communication technologies owned by

US companies. Thus, in 1919, the Radio Corporation of America, or RCA, was established. It acquired rights to the Marconi patents for the North American market and cross-licensed them for other major US patent portfolios. Fourteen US patent owners became shareholders of RCA. As a result, the patent deadlock was resolved and RCA, as well as the cross-licensed companies, quickly resumed commercialization of radio products in the US.

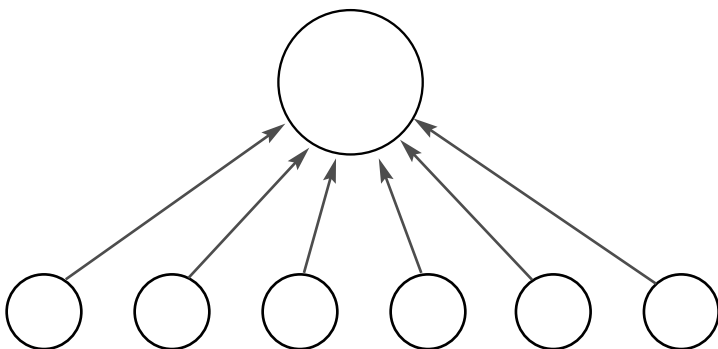
As Grindley and Teece report, RCA's cross-license model applied to the radio patent deadlock became a model for the future. Followed by AT&T's broad licensing practices, it established the basis for the cross-licensing practices currently used in the semiconductor industry. Regardless of an initial motive, be it a desire to increase royalty revenue or to secure freedom of action, or both, cross licensing seems to have positive socioeconomic implications. In a nutshell, it is not only an exchange of existing technologies, but also an exchange of future innovation possibilities.

The carrot model

Before the 1990s, many companies in the semiconductor industry generally kept technical know-how and other IPs in-house as a way of maintaining competitive advantage. However, this outlook changed as companies began to reach out to potential new markets and pushed to capitalize on the value of their IPs before the usefulness ran out. In quickly evolving technologies, many companies make their IPs available to other firms. These other potential licensees come from different industrial sectors and approach the dominant company—the potential licensor—to gain the rights to its IP. Such an approach, called the carrot model, is depicted in **Figure 3**.

The potential licensees are interested in licensing certain aspects of the company's IP because it meets their needs in a fast-growing industry or market. A company eager to develop technology faster than what its in-house R&D can achieve may look for IP from a licensor to help reach a fast-moving target market quickly.

Figure 3
The carrot model



From the perspective of the licensor, this model ensures revenue streams through license fees, royalties, and services. When a licensee purchases the rights to use the IP, it pays the licensor a combination of upfront fees and royalties. As sales volume increases, the licensee continues to pay the royalty on the IP it purchased. The licensor may continue to gain revenue by providing services for the maintenance of a particular IP. The long-term benefits include speeding up R&D cycles by channeling licensing revenue into new R&D activities, as IBM has done. Moreover, given that the licensee often comes from a different industry with which the licensor is not familiar, this model offers the licensor an opportunity to value the IP at its market value, thus leading to the optimization of its IP portfolio.

From the perspective of the licensee, the carrot model lowers the entry barriers to a new market. The firm can move in with new products clad in IP developed by a major player that has already met environmental or other regulatory standards. Ultimately, the process of measuring IP value and coming to an agreement is the key issue in this model. The licensor is more concerned about ensuring the accuracy of its revenue stream while maintaining a good relationship with the licensees. It can mitigate both technology and market risks by allowing a third party to manufacture and/or distribute in a foreign country. The licensee benefits by extracting value from the license and gaining subsequent lawsuit indemnification. However, the licensor must be cognizant of possible drawbacks, such as the potential IP value dilution, a negative impact on sales, and a loss of competitive advantage.

● The case of Motorola

Although Motorola has recently streamlined operations to focus on its core businesses, its technology portfolio is a result of vertically integrated structures. The company's operations historically extended from semiconductor design, through advanced automated test equipment and development of energy products such as batteries and new technology fuel cells for portable devices, to the design and manufacture of cellular phones. In the past, Motorola used its IP assets to maintain technological competitive advantage and to secure freedom of action, consistent with the stick and defense models.

However, the *Financial Times* (Acey 2002) reported that this approach has changed. Senior executives announced that Motorola was beginning to evaluate, on a case-by-case basis, which technologies were not being fully exploited and could be licensed out. The company is currently building business cases within business units to look at IP assets in order to aggressively maximize the return on those assets.

Motorola subsequently announced that it would make its entire 2.5G and 3G mobile phone platform technology available for licensing by third parties. Recent licensing agreements with Benq and Siemens are signs that Motorola has begun implementing this strategy. Capitalizing on many of its communication technologies, the company now welcomes opportunities to license its IP and return cash flow to the bottom line without further compromising product cost. These technology assets had already been amortized or were considered sunk costs as part of the R&D expense.

In another recent example, Motorola teamed with Agere Systems and Infineon in June 2002 to establish StarCore, a new company focused on developing and marketing digital signal processors (DSPs). "The central goal of the new StarCore," said its CEO, Thomas Lantzsch, "is to proliferate world-class DSP core technologies through open licensing." This move indicates that to execute its licensing strategy, Motorola applies a broad spectrum of licensing models. As a shareholder of StarCore, Motorola's strategy is to benefit through application of the market model, which we will describe later.

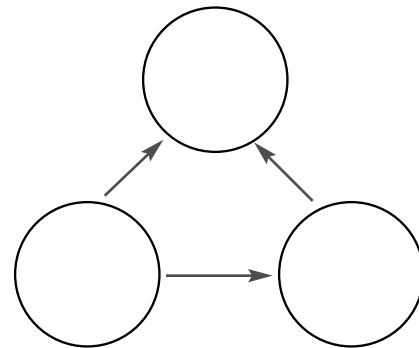
The consortium model

In an emerging market, standards—product size, type of raw materials, component interconnectivity, product technology, and so on—are often lacking. Standards enable economies of scale and cost-sensitive manufacturing by distributing the cost of production tools over a large number of competing companies. Often, IP issues are involved in these standards, and when the task of proliferating one type of industry standard becomes too great for one company, several companies form a consortium and work cooperatively to advance that standard (see **Figure 4**). The resulting IP is owned by one or several companies, and all the others become licensees as part of the agreement to proliferate the standard.

The Bluetooth Special Interest Group (SIG) is a good example of this model. It is a trade association comprising major players in the telecommunications, computing, and network industries that drives the development of low-cost short-range wireless specifications for mobile product connectivity. Its promoters, or standards owners, include 3Com, Agere, Rescission, IBM, Intel, Microsoft, Motorola, Nokia, and Toshiba. Other companies can join either as associates or as adopting members and can use the standard on a royalty-free basis.

Although the legal mechanisms vary from consortium to consortium, the key arrangement is that the parties involved contribute technology for developing standards and agree to grant all parties—members and non-members alike—a license to implement the standards on reasonable and non-discriminatory terms. All the consortia we reviewed shared the same principles, but the level of

Figure 4
The consortium model



cooperation and technology sharing varied, from joint development arrangements based on JV-type contracts, to a more loose arrangement in which member companies may freely join or leave the consortium without negative financial implications. The level of royalties also varied, from being royalty-free, to being paid among member companies based on IPs brought to the consortium, to being paid by standard adopters.

One of the notable consequences is that the model creates large markets for products and services. Some industry experts have predicted that the use of Bluetooth SIG wireless technology will result in cumulative incremental revenues of more than \$10 billion in the next few years. The bulk of these revenues will result from use of the technology to link mobile handsets with notebook computers and personal digital assistants (PDAs) to access data services, including e-mail. In addition, companies involved in creating standards can accrue large financial benefits by getting a technology head start and earning revenue from IP ownership and licensing. Of course, consumers benefit by receiving needed services at lower prices.

It is important to note that the standard creation processes using the consortium model may not always be beneficial to a company with a dominant market or breakthrough technology. They can be time-consuming and costly and require compromises that a member company may not be able to handle.

● The case of JEDEC

The Joint Electron Device Engineering Council is the semiconductor engineering standardization consortium, which is part of the Electronic Industries Alliance. Created in 1960 to coordinate industry efforts for the standardization of semiconductor devices, JEDEC has about 300 member companies. Membership on any of its committees is open to companies with a manufacturing presence in the United States.

JEDEC has well established the practices related to IP rights in setting standards. Committees must avoid drafting standards that would cover known patents. If they cannot avoid it, the member company holding the patented technology must make available a license to standard adopters either without compensation or under reasonable terms and conditions.

Formal and informal knowledge exchange is the key benefit to JEDEC members. The consortium's official website (www.jedec.org) emphasizes the knowledge sharing aspect:

Both large and small companies as well as users bring a mix of views representing a wide facet of this industry. The reason to become a member is to find out what the trends in a particular product area are and to find out the views of other companies as well as to mold and shape the final standards that JEDEC publishes....The JEDEC activity is where competitors who normally do not communicate get to talk about important industry trends.

One very visible contribution of JEDEC has been to set standards on the physical outlines of semiconductor chips. The standards accelerated specialization within the industry value chain as well as the growth of subcontractors specialized in printed circuit board assembly. Moreover, they allowed assembly houses to obtain economies of scale, increase returns on the capital invested in assembly and test tools, and enable the whole new industry of electronic design automation (EDA).

The market model

The market model, shown in **Figure 5**, applies to a situation in which a minor firm specializing in a narrow segment of a value chain captures a dominant technology position. When this happens, many major companies may approach the firm to license its technology, desiring to take advantage of the innovation and not fall behind their competitors in the development of next-generation products. The firm may first try to manufacture the product that embeds the technology. But because it does not have a broad enough market reach or sufficient capital, it ends up licensing the technology to other dominant players that do. This model may also benefit smaller players who have the opportunity to license the same technology and change the competitive landscape.

The financial gains enjoyed by the licensor are tremendous. As its technology proliferates and becomes the industry standard, its licensee base expands and revenue sources multiply. A former cost center focused on R&D can now be considered a profit center, or even spin off a new company. Once the licensees apply the technology to their products, the switching costs to another technology become so high that the continued revenue stream is virtually locked in.

The market model is fundamentally different from the carrot model on two key accounts. Its licensor is typically an emerging entity, rather than an established player in the industry. And IP licensing is a central rather than secondary strategy.

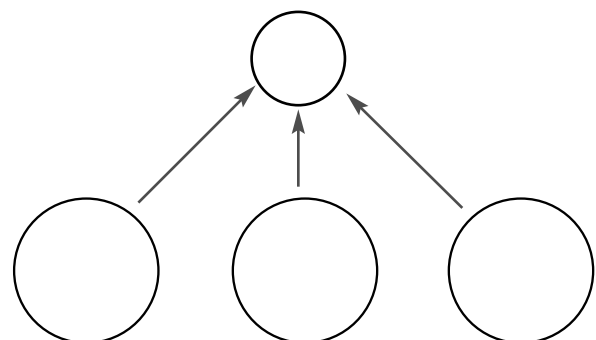
● The case of ARM

ARM spun off from Acron Computer Group as a joint venture between Apple, VLSI, and Acron. At the time, Acron was having difficulty finding an appropriate 16-bit microprocessor architecture for its computers. Constrained by financial and human resources, 12 engineers set out to develop a very simple microprocessor. Thus, the foundation was laid for Reduced Instruction Set Computer (RISC), despite the company's limited market base.

Soon after signing on VLSI Corporation as its first licensee, ARM began to expand into other industries. In 1993, TI integrated ARM technology into its Thumb Digital Processing core within two years of the start of their licensing agreement. This event signaled to other market participants ARM's expanding reach into end equipment markets outside the computing field. Today, Intel's Strong ARM processor is considered the de facto standard in the Pocket PC PDA. Even market leader Palm began to license the ARM platform. Palm also committed to TI's OMAP microprocessor platform, enabling full multimedia and wireless services. OMAP, which stands for Open Multimedia Application Platform, contains one of the successively developed ARM cores, synthesized with TI's own digital signal processor on a single chip.

Because of the central role of the microprocessor in most electronic applications, ARM has participated in many diverse markets, thereby spreading the risk. The industries it is involved in include automotive (airbag control and safety), consumer entertainment (Game Boy, MP3 players), imaging (digital cameras), industrial control (motor control, utility metering), networking (System-on-a-Chip design for embedded Web servers), security (Smartcards),

Figure 5
The market model



and wireless. Today, ARM controls 75 percent of the market for IP cores.

To extract value from IP assets, companies should use a broad spectrum of licensing models. Many of the semiconductor licensors we studied have adopted flexible strategies using all or most of the models described here.

When managing the knowledge supply chain, it is very important for licensees to understand the complexity of various licensing relationships and manage them effectively, just as with well-established buyer/supplier relationships. Although effective knowledge supply chain management becomes critical to operations management, IP licensing responsibilities are frequently scattered among technical and legal functions and are not sufficiently integrated with the supply management function. As in the case of traditional material purchases, companies may leverage their purchasing power for licensing-in transactions. Moreover, know-how is traditionally kept in-house as competitive advantage becomes available on the market. Buying versus making decisions should be part of any R&D investment choice.

As vertically integrated companies become less integrated, businesses are emerging that compete on knowledge, especially those that supply intangible product components such as designs, embedded software, or software design tools. No matter where we turn in the landscape of supply chains, the issue of knowledge management has become ubiquitous. ○

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